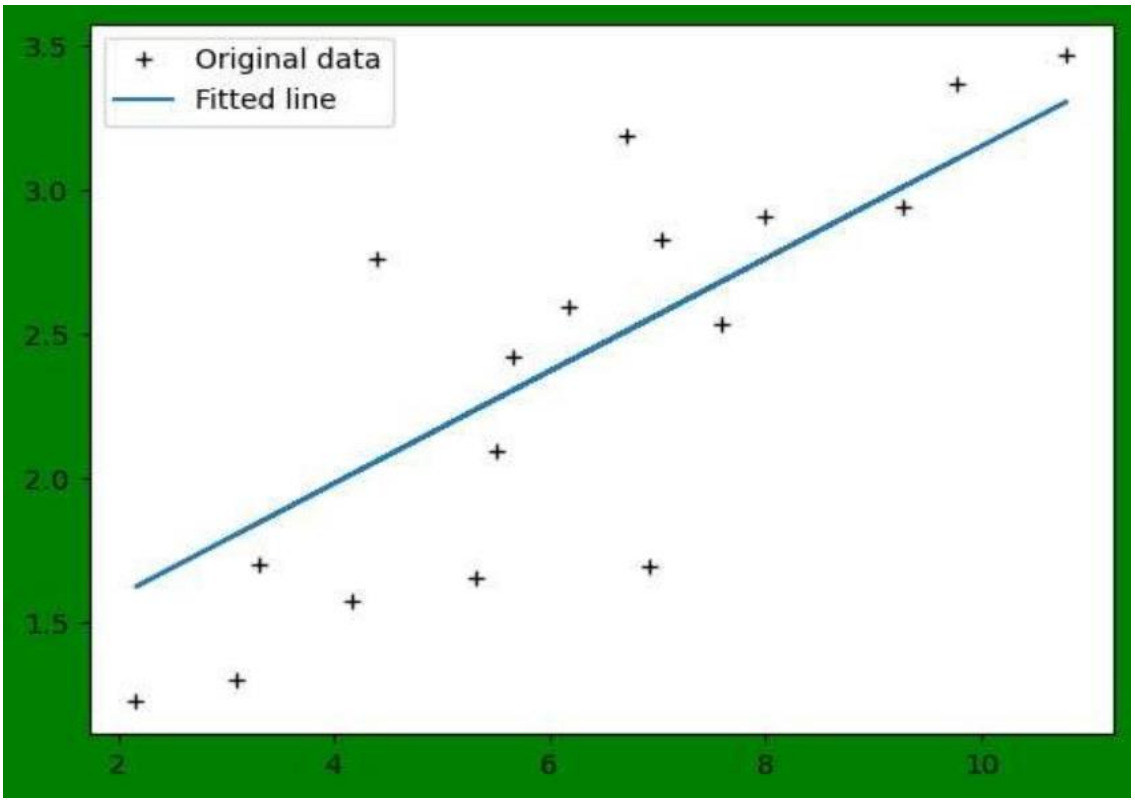
SID : 2534321

LAB1



LAB 2 :

```
Test Accuracy: 1.000
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 10)	350
dense_1 (Dense)	(None, 67)	737
dense_2 (Dense)	(None, 56)	3,808
dense_3 (Dense)	(None, 34)	1,938
dense_4 (Dense)	(None, 1)	35

Total params: 20,606 (80.50 KB)

Trainable params: 6,868 (26.83 KB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 13,738 (53.67 KB)

LAB : 3

Test accuracy: 0.8736000061035156

```
[27]: print(model.summary())
```

Model: "sequential"

Layer (type)	Output Shape	Param #
flatten (Flatten)	(None, 784)	0
dense (Dense)	(None, 56)	43,960
dense_1 (Dense)	(None, 45)	2,565
dense_2 (Dense)	(None, 32)	1,472
dense_3 (Dense)	(None, 24)	792
dense_4 (Dense)	(None, 10)	250

Total params: 49,039 (191.56 KB)

Trainable params: 49,039 (191.56 KB)

Non-trainable params: 0 (0.00 B)

None

LAB : 4

```
model = Sequential()

# Your code to add as many layers with activation functions of your choice:

model.add(Dense(200,activation='relu',input_shape=(n_features,)))
model.add(Dense(150,activation='relu',input_shape=(n_features,)))
model.add(Dense(100,activation='relu',input_shape=(n_features,)))
model.add(Dense(80,activation='relu',input_shape=(n_features,)))
model.add(Dense(40,activation='relu',input_shape=(n_features,)))


model.add(Dense(1))
```

Model: "sequential_2"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 200)	4,000
dense_1 (Dense)	(None, 150)	30,150
dense_2 (Dense)	(None, 100)	15,100
dense_3 (Dense)	(None, 80)	8,080
dense_4 (Dense)	(None, 40)	3,240
dense_5 (Dense)	(None, 1)	41

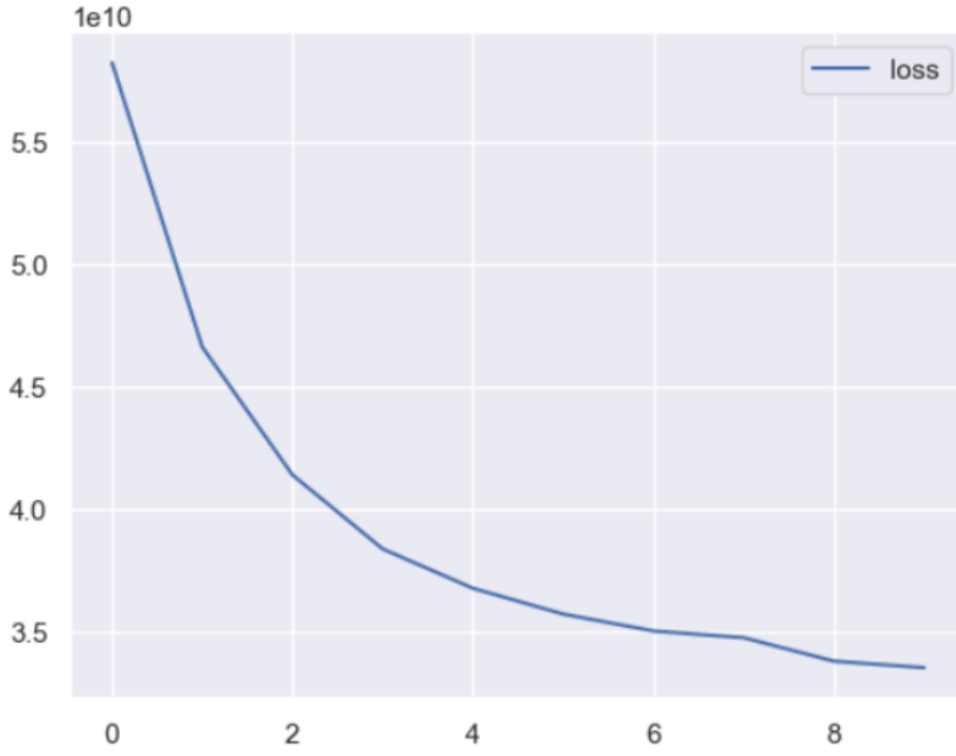
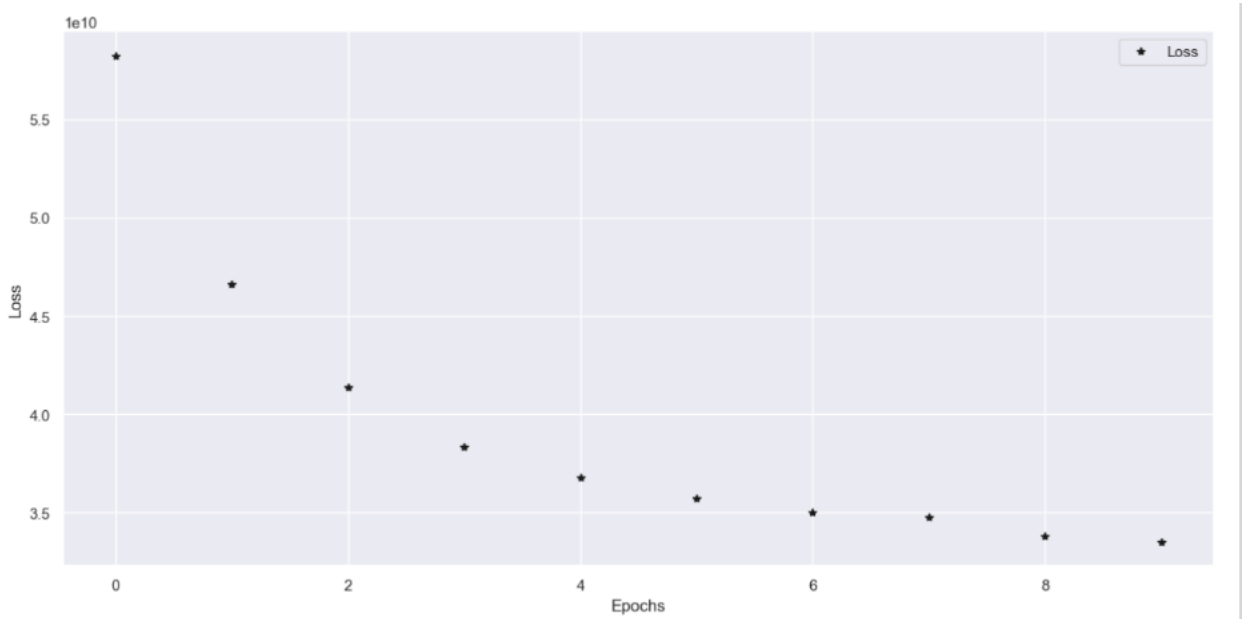
Total params: 60,611 (236.76 KB)

Trainable params: 60,611 (236.76 KB)

Non-trainable params: 0 (0.00 B)

None

5]:
```
# ! add your chosen optimiser !

model.compile(optimizer='Nadam', loss='mse')
```

]: <Axes: >